

A Balanced-Matched Double-Tuned Probe for *in Vivo* ^1H and ^{31}P NMR

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Since the advent of *in vivo* nuclear magnetic resonance surface-coil spectroscopy (1), there has been interest in using double-tuned surface coils to monitor more conveniently the dynamic changes in tissue metabolism. Nuclei that have been of most interest in spectroscopic studies thus far have been ^1H , ^{19}F , ^{31}P , ^{23}Na , and ^{13}C ; each of these nuclei provides information that is unique to itself. For example, ^1H studies have already shown that changes in lactate levels can be monitored *in vivo*, while ^{31}P is useful for observing changes in high-energy phosphates and pH. Using a double-tuned coil would enable one to monitor both nuclei simultaneously (2) so that changes in lactate levels, pH and high-energy phosphates could be better correlated. Similarly, other nuclei could be simultaneously monitored to provide a more complete metabolic picture of the tissue than could be accomplished using a single-tuned coil. Another potential use for double-tuned coils would be for ^{13}C spectroscopy with simultaneous ^1H decoupling (or vice versa, ^1H observation with ^{13}C decoupling), so that one can follow the metabolism of labeled substrates from which one can derive information on the activity of metabolic pathways (3).

A variety of double-tuned probe circuits, most of which were designed for ^{13}C spectroscopy, have already been published (2-6). Some of these incorporate trap circuits which are appropriately placed to act as filters to provide isolation between the two channels (6). The filters make many of the circuits difficult to construct and also degrade the sensitivities of the two channels. A more recent scheme proposed by Schnall *et al.* (2) has the advantages of both simplicity of construction and high performance but at the expense of the loss of isolation. Fortunately, the use of double-tuned coils to observe nuclei in a time-shared manner does not require isolation between the two channels because one is never irradiating simultaneously at two different frequencies. We report here the improvement in coil performance of a balanced-matched variant of the double-tuned circuit suggested by Schnall *et al.* (2).

Optimization of hardware is important due to the inherent low sensitivity of the NMR experiment. Because of the high conductivity of biological tissues, losses in probe sensitivity and, consequently, signal must be minimized while performing *in vivo* studies (7, 8). The fact that one can compensate for the degradation of sensitivity caused by dielectric losses is illustrated by the successful attempts of others in reducing the dielectric losses induced by *in vivo* tissues on single-tuned probe circuits (9, 10).

These attempts have included the use of Faraday shields between the coil and sample (9) and modifications in the standard tuning and matching circuit, referred to as balanced matching (10). These losses are an important concern as seen by the significant increase in the *in vivo* signal-to-noise ratio (S/N) achieved by these schemes.

The balanced-matched, double-tuned probe circuit that we used is shown in Fig. 1A together with the "standard" matching double-tuned circuit (Fig. 1B) similar to that described by Schnall *et al.* (2). The sample coil, which has an inductance L_1 , is an 8×12 mm two-turn elliptical surface coil made of No. 18 gauge solid copper wire. C_1 , C_2 , C_3 , and C_4 are tunable capacitors (1–30 pF, Johanson, Boonton, NJ 07005). For the case of the single-tuned circuit, the coil can be tuned to one resonance frequency with the elements C_1 , and C_3 (5) (Fig. 1B, disregarding, for the moment, the trap circuit elements L_2 and C_2); C_1 is for tuning and C_3 is for impedance matching. In the balanced-matched scheme (Fig. 1A, disregarding, for the moment, the trap circuit elements L_2 and C_2), C_4 is added to balance the coil with respect to ground, thus reducing the losses incurred by loading the coil with a conductive sample (10). In this scheme, C_3 and C_4 become the matching capacitors. According to Schnall *et al.* (2), a coil can be made to resonate at two different frequencies by the introduction of a trap circuit with elements L_2 and C_2 (Figs. 1A, B). The inductor with inductance L_2 is optimized for the appropriate frequencies for which one wishes to double tune the coil to (i.e., ^{31}P – ^1H , ^{23}Na – ^1H , . . .). In our balanced-matched double-tuned probe circuit (Fig. 1A), an extra capacitor C_4 is introduced which allows us to tune and match both frequencies better and also reduces the dielectric losses when the coil is loaded (Table 1).

In vivo NMR spectroscopy was performed on our 5.6 T spectrometer (^1H and ^{31}P frequencies; 236.8 and 95.8 MHz) slightly modified with high- and low-pass filters, respectively, placed to protect the ^1H and ^{31}P preamplifiers. A live rat was anesthetized with 1.5% isoflurane, the scalp removed, and the coil placed directly over the exposed skull. For purposes of comparison, experimental conditions were kept the same as much as possible for both the balanced- and standard-matched coils (the same rat was used in both experiments). The water proton signal (observed through the ^1H channel

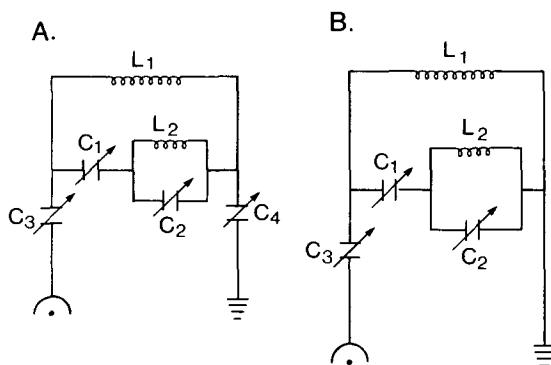


FIG. 1. (A) Balanced-matched and (B) standard double-tuned probe circuits. For single-tuned circuits, disregard the trap circuit elements L_2 and C_2 .

TABLE 1

Coil Performance of Balanced-Matched and Standard Double-Tuned Probes

Coil	Q (unloaded) ^a		90° pulse width ^b		Q (loaded) ^c		S/N (<i>in vivo</i>)	
	³¹ P	¹ H	³¹ P	¹ H	³¹ P	¹ H	³¹ P ^d	¹ H ^e
Standard	174	113	15.1	36.0	120	91	6.2	48.3
Balanced	213	189	9.2	17.3	192	169	9.5	78.2

^a $Q = 2 \times (\nu / \Delta\nu_{0.71})$; ν , resonance frequency; $\Delta\nu_{0.71}$, bandwidth at 0.71 peak height.^b 90° pulse width at the center of the coil measured with a 8.5% phosphoric acid microsphere phantom.^c Q estimated with the coil placed over a rat head.^d The ratio of the peak height of β -phosphate resonance of ATP to rms noise.^e The ratio of the peak height of $-\text{CH}_3$ resonance of *N*-acetyl aspartate to rms noise.

of the coil) was used for shimming; in both cases the water linewidth was shimmed to 0.13 ppm. The ¹H spectra of the rat brain were obtained by employing a presaturation-Hahn spin-echo pulse sequence (11). Two hundred free induction decays (FIDs) were collected with a sweep width of ± 2500 Hz, 4K data points, and a total repetition time of 2.5 s (Figs. 2A, C). For ³¹P spectroscopy, a selective saturation pulse sequence was employed to remove signals arising from bone and phospholipids (12). Acquisition parameters were the same as for the ¹H experiment except the sweep width was changed to ± 5000 Hz (Figs. 2B, D).

Table 1 lists the coil performance of both the balanced-matched and standard double-tuned probe circuits. The quality factor Q of the coil was measured using the reflected power detected from a Wavetek sweep generator and Tektronix oscilloscope, and was calculated by doubling the ratio of the resonance frequency to the bandwidth at 0.71 of the peak height (9, 13). The 90° pulse width in the center of the coil (measured with a 8.5% phosphoric acid microsphere phantom) was determined to compare the efficiencies of power delivery of the two probe configurations (2). The peak heights of the β -phosphate resonance of ATP and the $-\text{CH}_3$ resonance of *N*-acetyl aspartate (NAA) were used for calculation of the *in vivo* S/N for ³¹P and ¹H, respectively.

The data in Table 1 illustrate the effectiveness of the balanced-matched scheme in improving coil performance on the parameters measured. The Q of the balanced-matched coil, both loaded and unloaded, are better than that of the unbalanced coil. The difference in Q is reflected in the coil efficiency as seen in the decrease in 90° pulse widths from the standard configuration to the balanced configuration. Finally, and most importantly, there is a greater than 50% improvement in S/N *in vivo* for both the ¹H and ³¹P channels.

Previously, use of a double-tuned coil meant that one must tolerate losses in sensitivity in one or both channels. The double-tuned coil suggested by Schnall *et al.* (2) already eliminates many of the losses involved in the elements used in the construction of the coil. Our modification of this circuit minimizes the losses in sensitivity incurred during loading of the coil by the highly conductive samples that one encounters *in vivo*. The use of this type of double-tuned coil enables more efficient use of the NMR spectrometer by enabling one to observe two different nuclei simultaneously while

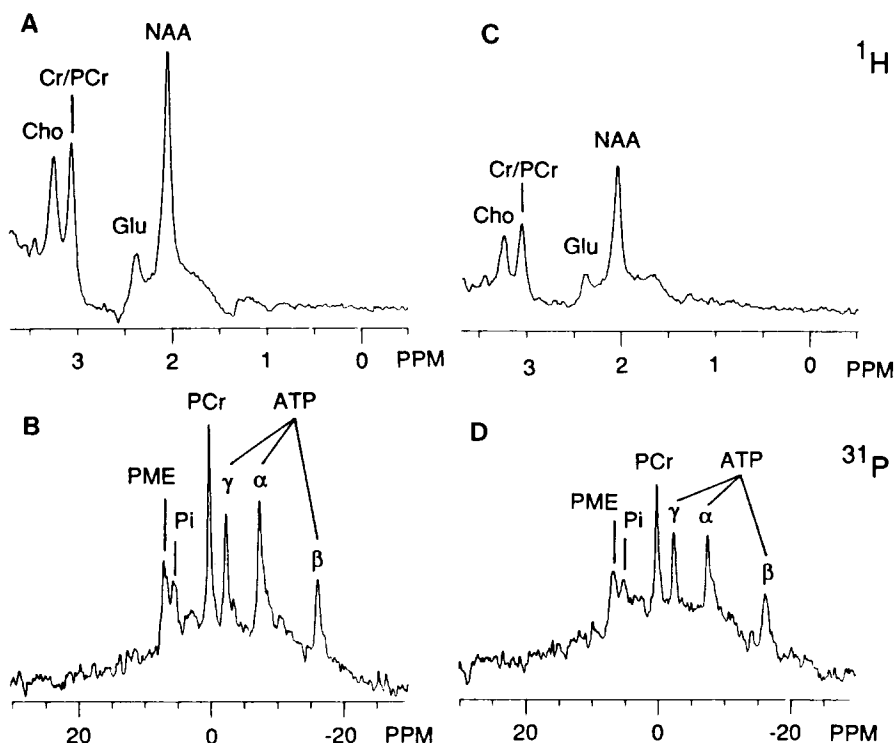


FIG. 2. *In vivo* NMR spectra of a rat brain. (A) ^1H spectrum obtained with the balanced-matched double-tuned coil, and sensitivity enhanced by multiplication of the FIDs by an exponential function of line broadening of 3 Hz before Fourier transform. (B) ^{31}P spectrum obtained with the balanced-matched double-tuned coil, and line broadening of 25 Hz. (C) ^1H and (D) ^{31}P spectra obtained with the standard double-tuned coil using the same spectral parameters as those of (A) and (B). Signal assignments are as follows: Cho, choline; Cr/PCr, creatine/creatine phosphate; Glu, glutamate; NAA, *N*-acetyl aspartate; PME, phosphomonoesters; Pi, inorganic phosphate; PCr, creatine phosphate; α -, β -, γ -ATP, α -, β -, γ -phosphates of adenosine triphosphate.

still maintaining good sensitivity in both channels. This can provide more complete information on the dynamic state of the tissue under study since complementary information can be obtained simultaneously from two different nuclei.

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